

# Appendix: Band-Limitedness of the Hardy Z-Function — Supplementary Material

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This document collects supplementary material for the main paper “Band-Limitedness of the Hardy Z-Function in the Phase Variable. Section, equation, and theorem references prefixed with extttMain: refer to the main paper; local references are internal to this document.

## A Answers to Frequently Anticipated Questions

This appendix collects short, colloquial explanations of terminology and conventions used in the main text.

### A.1 Why “inverse phase pullback” and not “phase pushforward”?

The reason is semantic, not technical. The intuition is geometric, in the plain English sense of the words.

*Pullback.* Picture yourself standing at a point with stuff scattered across a distant landscape. You throw out strings, hook onto the stuff that is out there, and reel it back in toward where you are standing — like hauling in a kite. The object ends up concentrated at your location, retrieved from the space it was spread over. That retrieving motion is a pullback.

*Pushforward.* Now picture yourself sitting with a kite wound up at your feet and you want it out in the sky: you let out the line and send it forth, expanding it into the space it will occupy. That outgoing, spreading motion is a pushforward.

Apply that picture here.  $Z$  is being sent outward: it is spread across the  $t$ -line, expanded into that space. To build  $Y$  on the  $u$ -line, one sits at a point  $u$ , reaches back through the phase map via  $\Theta^{-1}$ , grabs the value  $Z(\Theta^{-1}(u))$ , and reels it in. That retrieving is exactly why the operation is called a pullback:  $Z$  has been sent out, and one has to pull it back to gather it at  $u$ . Since the reeling is done along  $\Theta^{-1}$ , the qualifier is “inverse phase.” The adjective “unitary” records that the half-Jacobian  $1/\sqrt{\Theta'(\Theta^{-1}(u))}$  is included so that  $L^2$ -mass is preserved under the operation.

**Contravariance and covariance.** The two directions carry standard names in differential geometry: a pullback is *contravariant* and a pushforward is *covariant*, referring to how each transforms under composition of maps. Objects that live as functions or scalar fields — things one evaluates at a point — transform contravariantly: they pull back, against the direction of the map. Objects that live as tangent vectors, velocities, or measures — things that carry a notion of flow or mass — transform covariantly: they push forward, with the direction of the map. The split is the same one that distinguishes covectors from vectors, or differential forms from densities: it records which kind of geometric object is being transported and which way the arrow of transport naturally runs.

Under this classification,  $Z$  and  $Y$  are scalar-like, half-density data, and the natural transport between them is contravariant — a pullback.

**A remark on the dual naming.** Technically one could reverse the point of view and call  $Z$  itself the *phase pushforward* of the stationary process  $Y$ : starting from  $Y$  on the  $u$ -line, one lets out the line along  $\Theta$  and expands  $Y$  outward into  $t$ -space, obtaining  $Z$ . Equivalently, the stationary process  $Y$  could be defined as the pushforward of  $Z$  along the phase — sending  $Z$  forth into  $u$ -coordinates. This naming is formally consistent, but it reads against the grain of what is being constructed. Stationarity is a property of something gathered and settled; producing it by *pushing* an object outward into a new space is the wrong English motion. One naturally assembles a stationary object by *pulling in* values from the non-stationary source. “Unitary inverse phase pullback” is therefore the name used throughout: it matches the geometric direction of the construction and the linguistic image of stationarity as something retrieved, not something broadcast.

**Ditty.** English grammar and category theory disagree here. In plain English a *function* is the active thing — something that acts, takes an argument, produces a value; a *measure* is what remains, passively assigning mass to sets without acting on anything. The English grammar would therefore pair functions with pushforward and measures with pullback. Category theory goes the other way: functions transform contravariantly (they pull back), measures transform covariantly (they push forth), because a pullback of a function  $f$  along  $\Theta$  is the composition  $f \circ \Theta$  (the function retreats along the map), while the pushforward of a measure  $\mu$  is  $\mu \circ \Theta^{-1}$  (the mass is transported forward along the map). The category-theoretic convention wins in practice; the English reading is the anti-pattern. The mnemonic records the inversion:

In English a function acts, a measure stays put;  
in category theory that reading’s kaput:  
functions pull back, measures push forth,  
and the unitary inverse phase pullback points north.

## A.2 Why is the spectral measure non-negative?

The autocorrelation of any real, locally square-integrable function is a positive-definite tempered distribution: the Gram matrix of finitely many translates of  $Y$  restricted to a common window has non-negative eigenvalues, a one-line consequence of Cauchy–Schwarz. Bochner–Schwartz identifies the Fourier transform of a positive-definite tempered distribution with a non-negative tempered measure, and Wiener–Khinchin identifies that Fourier transform with  $S = |\hat{Y}|^2$ . Non-negativity of the spectral measure is the Fourier-side restatement of this universal property of autocorrelations; the support condition  $\text{supp } S \subseteq [-1, 1]$  is delivered by Theorem ?? and combines with non-negativity to invoke Akhiezer.

## A.3 What is being autocorrelated?

The autocorrelation in Lemma ?? is that of  $Y$  itself, the unitary inverse phase pullback of  $Z$ . In  $u$ -coordinates,

$$C(\eta) = \lim_{U \rightarrow \infty} \frac{1}{2U} \int_{-U}^U Y(u) Y(u + \eta) du \quad (1)$$

a positive-definite tempered distribution. Because the map  $Z \mapsto Y$  is an  $L^2$ -isometry under  $u = \Theta(t)$ , this is equivalent to correlating  $Z$  against itself with respect to the  $u$ -shift  $\eta$ , which is a nonlinear (roughly logarithmic) shift in the original  $t$ -coordinate.

## A.4 What does the picture look like?

Three objects merit mental plots.

*The function  $Y(u)$ .* An envelope-modulated real oscillation of zero mean. The envelope decays like  $1/\sqrt{\log t}$  expressed in  $u$ -coordinates. The zero crossings are the images under  $\Theta$  of the non-trivial Riemann zeros; the phase variable straightens out their spacing.

*The autocorrelation  $C(\eta)$ .* A sharp peak at  $\eta = 0$ , symmetric in  $\eta$  because  $Y$  is real, with oscillatory decay whose frequency content sits inside  $[-1, 1]$ .

*The spectral measure  $S$ .* A non-negative tempered measure strictly supported in  $[-1, 1]$ , symmetric about  $\xi = 0$  after the even extension. Its shape inside  $[-1, 1]$  encodes the distribution of zero spacings of  $Z$  in  $u$ -coordinates.

## A.5 Locally $L^2$ and the growing variance

$Y$  is locally  $L^2$ , on compact subsets  $[-U, U]$  of arbitrarily large length  $U$ . The band-limit established in Theorem ?? is a statement about the Fourier support of  $Y$  as a tempered distribution on  $\mathbb{R}$ . All uses of Paley–Wiener, Bochner–Schwartz, and Akhiezer in the main text are read in the tempered-distributional form, the framework for band-limited objects that are not globally  $L^2$ .

The growing variance manifests in three compatible ways.

*Envelope in  $u$ -coordinates.*  $Y(u)$  has an amplitude envelope that grows slowly as  $|u| \rightarrow \infty$ ; the local mean-square  $\bar{P}(U) := \frac{1}{2U} \int_{-U}^U Y(u)^2 du$  grows without bound as  $U \rightarrow \infty$ .

*Autocorrelation as a tempered distribution.*  $C(\eta)$  is the windowed-limit positive-definite distribution of §A.3. Bochner–Schwartz identifies its Fourier transform with the non-negative tempered measure  $S$ , and Theorem ?? localizes  $S$  to  $[-1, 1]$ .

*Spectral measure as a tempered measure on  $[-1, 1]$ .*  $S$  is a non-negative tempered measure on  $\mathbb{R}$ , supported in  $[-1, 1]$ , of infinite total mass.

**Representation at finite window.** Pointwise values of the spectral density at interior  $\xi_0 \in (-1, 1)$  are approached through windowed periodograms

$$P_U(\xi_0) := \frac{1}{2U} \left| \int_{-U}^U Y(u) e^{-i\xi_0 u} du \right|^2 \quad (2)$$

Any figure produced at finite  $U$  is a *representation* at window scale  $U$ .

## A.6 Scale of the spectral representation at finite window

Fix  $\xi \in (-1, 1)$  and let  $P_U(\xi)$  be the windowed periodogram. Two features of the plot change with  $U$ : the overall vertical scale, and the local fluctuation amplitude around that scale. Both are controlled by the local mean power

$$\bar{P}(U) := \frac{1}{2U} \int_{-U}^U Y(u)^2 du = \frac{1}{2U} \int_{\Theta^{-1}(-U)}^{\Theta^{-1}(U)} Z(t)^2 dt \quad (3)$$

the second equality by the  $L^2$ -isometry of the unitary inverse phase pullback. By Parseval in the tempered sense,

$$\int_{-\infty}^{\infty} P_U(\xi) d\xi = 2\pi \bar{P}(U) \quad (4)$$

Using  $\Theta(t) \sim \frac{t}{2} \log(t/(2\pi e)) + ct$  and the Hardy–Littlewood moment  $\int_0^T Z(t)^2 dt \sim T \log T$  ([11], §7.3),

$$\bar{P}(U) \sim \log(\Theta^{-1}(U)) \sim \log U + \log \log U + O(1) \quad (U \rightarrow \infty) \quad (5)$$

$P_U(\xi)$  fluctuates around  $\bar{P}(U)$  on a scale comparable to  $\bar{P}(U)$  itself, with wider fluctuations near the edges  $|\xi| \rightarrow 1^-$ .

## A.7 The normalized spectral measure

At every finite  $U$  the periodogram  $P_U$  has finite total mass  $2\pi\bar{P}(U)$  by Parseval. Dividing by this total mass produces the normalized spectral probability measure at scale  $U$ ,

$$\tilde{P}_U(\xi) := \frac{P_U(\xi)}{2\pi\bar{P}(U)}, \quad \int_{-\infty}^{\infty} \tilde{P}_U(\xi) d\xi = 1 \quad (6)$$

a probability density on  $\mathbb{R}$  at every  $U$ . By Theorem ?? the probability mass of  $\tilde{P}_U$  outside  $[-1, 1]$  vanishes as  $U \rightarrow \infty$ , so the family is tight on  $[-1, 1]$ . By Helly–Prokhorov,  $\tilde{P}_U$  converges weakly as  $U \rightarrow \infty$  to a probability measure  $\nu_\infty$  on  $[-1, 1]$ :

$$\int f d\tilde{P}_U \longrightarrow \int f d\nu_\infty \quad \text{for every } f \in C_b(\mathbb{R}) \quad (7)$$

$\nu_\infty$  is the normalized spectral density of  $Y$ : a probability measure of total mass 1 supported in  $[-1, 1]$ .

$S$  and  $\nu_\infty$  are distinct, both well-defined:

- $S$ : non-negative tempered spectral measure on  $[-1, 1]$ , infinite total mass, delivered by Theorem ?? and Bochner–Schwartz ([9], Theorem 7.7).
- $\nu_\infty$ : probability measure on  $[-1, 1]$ , the weak-\* limit of the normalized windowed periodograms  $\tilde{P}_U$ .

## A.8 Cramér representation of the sample path

As a generalized second-order stationary process on  $\mathbb{R}_u$  with non-negative tempered spectral measure  $S$  supported in  $[-1, 1]$ ,  $Y$  admits a Cramér spectral representation of its sample path,

$$Y(u) = \int_{-1}^1 e^{i\xi u} d\Phi(\xi) \quad (8)$$

where  $d\Phi$  is a complex-valued orthogonal Borel measure on  $[-1, 1]$  with covariance structure

$$\mathbb{E}[d\Phi(\xi) \overline{d\Phi(\xi')}] = \delta(\xi - \xi') dS(\xi) \quad (9)$$

The support condition  $\text{supp } S \subseteq [-1, 1]$  of Theorem ?? localizes  $d\Phi$  to the interval  $[-1, 1]$ . Reality of  $Y$  on  $\mathbb{R}$  is equivalent to the Hermitian symmetry  $d\Phi(-\xi) = \overline{d\Phi(\xi)}$ , under which  $S$  is an even measure on  $[-1, 1]$ . Test functions  $f$  are paired with  $d\Phi$  under the  $L^2(S)$  isometry

$$\mathbb{E} \left| \int f(\xi) d\Phi(\xi) \right|^2 = \int |f(\xi)|^2 dS(\xi) \quad (10)$$

which requires  $f \in L^2(S)$  and permits  $S$  to have infinite total mass. The Cramér representation exists at the level of  $S$ : it is the reconstruction theorem for the process.  $\nu_\infty$  is the scale-invariant spectral shape of the same process.

**Inverse representation of  $d\Phi$ .** The orthogonal Borel measure  $d\Phi$  is recovered from the sample path  $Y$  by the truncated Fourier inversion

$$\Phi(B) = \lim_{U \rightarrow \infty} \frac{1}{2\pi} \int_{-U}^U Y(u) \widehat{\mathcal{K}}_B(-u) du \quad (11)$$

for every Borel set  $B \subseteq [-1, 1]$ , the limit taken in  $L^2(\Omega_P, \mathbb{P})$ ; here  $\widehat{\mathcal{K}}_B(u) = \int_B e^{i\xi u} d\xi$ . Equivalently, for every  $f \in L^2(S)$ ,

$$\int_{-1}^1 f(\xi) d\Phi(\xi) = \lim_{U \rightarrow \infty} \frac{1}{2\pi} \int_{-U}^U Y(u) \check{f}(u) du, \quad \check{f}(u) = \int_{-1}^1 f(\xi) e^{-i\xi u} d\xi \quad (12)$$

again in  $L^2(\Omega_P, \mathbb{P})$ . Existence of the limit is the  $L^2(S)$  isometry (10) together with the truncated Fourier inversion on  $Y$ .

**Inserting  $d\Phi$  into the forward representation.** Fix admissible  $c$  and  $t \in \mathbb{R}$ . Apply (12) with test function  $f(\xi) = e^{i\xi\Theta(t)}$ ; this  $f$  is continuous on  $[-1, 1]$  hence in  $L^2(S|_{[-1, 1]})$ . The inner  $\xi$ -integral in the inverse formula is Lebesgue,

$$\check{f}(u) = \int_{-1}^1 e^{i\xi\Theta(t)} e^{-i\xi u} d\xi = \int_{-1}^1 e^{i\xi(\Theta(t)-u)} d\xi = \frac{2 \sin(\Theta(t) - u)}{\Theta(t) - u} \quad (13)$$

so that (12) reads

$$\int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi) = \lim_{U \rightarrow \infty} \frac{1}{\pi} \int_{-U}^U Y(u) \frac{\sin(\Theta(t) - u)}{\Theta(t) - u} du \quad (14)$$

in  $L^2(\Omega_P, \mathbb{P})$ . The passage from the nested  $(\xi, u)$ -form of (12) to the single  $u$ -integral (14) exchanges the order of  $\xi$ - and  $u$ -integration. At every finite  $U$  this is Fubini on the product measure  $d\xi \otimes du$  on  $[-1, 1] \times [-U, U]$ : the integrand satisfies

$$|Y(u) e^{i\xi(\Theta(t)-u)}| = |Y(u)| \quad (15)$$

and

$$\int_{-U}^U \int_{-1}^1 |Y(u)| d\xi du = 2 \int_{-U}^U |Y(u)| du < \infty \quad (16)$$

since  $Y$  is locally  $L^2$  on  $\mathbb{R}$  hence locally  $L^1$  on the compact  $[-U, U]$ ;  $[-1, 1]$  has finite Lebesgue measure. Fubini therefore applies: the double integral factors in either order, with the same value,

$$\int_{-U}^U Y(u) \left( \int_{-1}^1 e^{i\xi(\Theta(t)-u)} d\xi \right) du = \int_{-1}^1 \left( \int_{-U}^U Y(u) e^{i\xi(\Theta(t)-u)} du \right) d\xi \quad (17)$$

and the  $L^2(\Omega_P, \mathbb{P})$ -limit  $U \rightarrow \infty$  on both sides exists, with the limit on the right equal to  $\int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi)$  by truncated Fourier inversion for  $Y$ . The bound (15) is independent of  $c$ , and the only appearance of  $c$  in (17) is through the real translate  $\Theta(t) = \theta(t) + ct$  in the phase; Fubini's hypothesis is therefore satisfied for every admissible  $c$ .

**Time-warped representation of  $Z$ .** The limit on the right of (14) is the sinc-kernel sampling of  $Y$  at the point  $\Theta(t) \in \mathbb{R}$ . Since  $Y$  is band-limited to  $[-1, 1]$  by Theorem ?? and  $\Theta(t) \in \mathbb{R}$ , Paley–Wiener sampling gives

$$\lim_{U \rightarrow \infty} \frac{1}{\pi} \int_{-U}^U Y(u) \frac{\sin(\Theta(t) - u)}{\Theta(t) - u} du = Y(\Theta(t)) \quad (18)$$

in  $L^2(\Omega_P, \mathbb{P})$ . Combining (14), (18), and the real-line identity (??) ( $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$ ),

$$Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t)) = \sqrt{\Theta'(t)} \int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi) \quad (19)$$

for every admissible  $c$  and every  $t \in \mathbb{R}$ . This is the unitary time-change of the Cramér representation: the same orthogonal Borel measure  $d\Phi$  that represents the stationary sample path  $Y$  represents  $Z$  after the reparametrisation  $u \mapsto \Theta(t)$  and the half-Jacobian amplitude  $\sqrt{\Theta'(t)}$ . The representation (19) depends on  $c$  only through  $\Theta = \theta + c \cdot \text{id}$ ; the orthogonal Borel measure  $d\Phi$  itself is a spectral object of  $Y$ , independent of  $c$ .

**Covariance of  $Z$ .** From (19), the covariance of  $Z$  at  $t, s \in \mathbb{R}$  is

$$R_Z(t, s) := \mathbb{E}[Z(t) \overline{Z(s)}] = \sqrt{\Theta'(t) \Theta'(s)} \int_{-1}^1 e^{i\xi(\Theta(t) - \Theta(s))} dS(\xi) \quad (20)$$

*Derivation of (20) from (19).* Fix  $t, s \in \mathbb{R}$ . Expanding the product and pulling the amplitudes outside the expectation,

$$\mathbb{E}[Z(t) \overline{Z(s)}] = \sqrt{\Theta'(t) \Theta'(s)} \mathbb{E} \left[ \int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi) \overline{\int_{-1}^1 e^{i\xi'\Theta(s)} d\Phi(\xi')} \right] \quad (21)$$

The two Wiener integrals are each elements of  $L^2(\Omega_P, \mathbb{P})$  with  $L^2(S)$ -norms equal to  $\|e^{i\cdot\Theta(t)}\|_{L^2(S)} = S([-1, 1])^{1/2}$  and  $\|e^{i\cdot\Theta(s)}\|_{L^2(S)} = S([-1, 1])^{1/2}$ , which are finite whenever the integrals are taken against a compactly supported test function — here the restriction of  $e^{i\cdot\Theta(t)}$  and  $e^{i\cdot\Theta(s)}$  to the compact set  $[-1, 1]$ . The Wiener–Itô product rule (the bilinear form of (10)) gives

$$\mathbb{E} \left[ \int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi) \overline{\int_{-1}^1 e^{i\xi'\Theta(s)} d\Phi(\xi')} \right] = \int_{-1}^1 e^{i\xi\Theta(t)} \overline{e^{i\xi\Theta(s)}} dS(\xi) \quad (22)$$

Combining (21) and (22) gives (20).

The exchange of the two  $\xi$ -integrals with the expectation in (22) is an instance of the bilinear Itô isometry and not an application of Fubini directly. The Fubini justification of that bilinear isometry is standard: for simple functions  $f = \sum_k a_k \mathbb{1}_{B_k}$  with  $B_k \subseteq [-1, 1]$  pairwise disjoint and Borel, and  $g = \sum_\ell b_\ell \mathbb{1}_{C_\ell}$  similarly, orthogonality of the increments  $\{\Phi(B_k)\}$  gives

$$\mathbb{E} \left[ \int f d\Phi \overline{\int g d\Phi} \right] = \sum_k \sum_\ell a_k \overline{b_\ell} \mathbb{E}[\Phi(B_k) \overline{\Phi(C_\ell)}] = \sum_k a_k \overline{b_k} S(B_k \cap C_k) = \int f \overline{g} dS \quad (23)$$

where in the middle equality we used  $\mathbb{E}[\Phi(B_k) \overline{\Phi(C_\ell)}] = S(B_k \cap C_\ell)$  (the integrated form of (9)) and the fact that for simple functions the double sum is finite so Fubini applies pointwise. Bounded-continuous functions on  $[-1, 1]$  are  $L^2(S)$ -limits of simple functions (standard, since  $S([-1, 1]) < \infty$ )

if one replaces  $S$  by  $S$ -on- $[-1, 1]$ , which is finite because Theorem ?? supports  $S$  in  $[-1, 1]$  but does not bound its total mass; for the identity (22) the tests  $e^{i\Theta(t)}$  and  $e^{i\Theta(s)}$  are bounded on the compact set  $[-1, 1]$ , so they are in  $L^2(S|_{[-1, 1]})$  even when  $S$  itself is infinite on  $\mathbb{R}$ , and the bilinear form in (22) extends continuously in both arguments by the  $L^2(S)$  isometry (10), yielding the claim. No assumption on  $c$  is used beyond admissibility, which is the standing hypothesis on  $\Theta$ .

The exchange in (19) itself — the equality  $\sqrt{\Theta'(t)}Y(\Theta(t)) = \sqrt{\Theta'(t)}\int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi)$  — is the pointwise substitution  $u \mapsto \Theta(t)$  in (8), valid because  $\Theta(t) \in \mathbb{R}$  for  $t \in \mathbb{R}$  and the sample path  $Y(\Theta(t))$  is defined by the Cramér representation evaluated at the point  $\Theta(t)$ . No integral exchange is required: the inner Wiener integral is the same element of  $L^2(\Omega_P, \mathbb{P})$ , with the phase  $e^{i\xi u}$  evaluated at  $u = \Theta(t)$ .  $\square$

The identity (20) is the time-warped autocovariance of the stationary underlying process. Non-stationarity of  $Z$  is expressed entirely through the pair  $(\Theta'(t)\Theta'(s))^{1/2}$  and the argument  $\Theta(t) - \Theta(s)$ : the frequency content is the same  $S$  supported in  $[-1, 1]$ ; the time dependence is absorbed into the biholomorphic reparametrisation  $\Theta$  and the half-Jacobian amplitude.

## A.9 Variance structure function

At coincident times  $s = t$  the covariance (20) reduces to the variance of  $Z(t)$ , and the  $\xi$ -integral in (20) loses its oscillatory factor. Write

$$\sigma_Y^2 := S([-1, 1]) = \int_{-1}^1 dS(\xi) \quad (24)$$

for the total mass of  $S$  restricted to  $[-1, 1]$ ; this quantity is finite when  $Y$  is second-order on  $\mathbb{R}$ , which is the hypothesis of the Cramér representation (8). Setting  $s = t$  in (20) then gives  $\Theta(t) - \Theta(s) = 0$ , the exponential equals 1, and

$$\text{Var } Z(t) = R_Z(t, t) = \Theta'(t) \int_{-1}^1 dS(\xi) = \sigma_Y^2 \Theta'(t) \quad (25)$$

for every  $t \in \mathbb{R}$ . The variance of  $Z$  is the total spectral mass of  $Y$  amplified by the instantaneous phase rate  $\Theta'(t)$ ; it is the diagonal restriction of the covariance, with no dependence on the off-diagonal phase.

**Pinning of  $\sigma_Y^2$  by the Hardy–Littlewood second moment.** The value  $\sigma_Y^2 = 2$  is not a free calibration; it is pinned as a theorem by the Hardy–Littlewood second moment of  $\zeta$ . Integrating (25) against  $dt$  on  $[0, T]$ ,

$$\int_0^T \text{Var } Z(t) dt = \sigma_Y^2 \int_0^T \Theta'(t) dt = \sigma_Y^2 [\Theta(T) - \Theta(0)]. \quad (26)$$

On the left, using  $|Z(t)| = |\zeta(\frac{1}{2} + it)|$  on  $\mathbb{R}$  (from  $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$  with  $|e^{i\theta(t)}| = 1$  for  $t \in \mathbb{R}$ ),

$$\int_0^T \text{Var } Z(t) dt = \int_0^T \mathbb{E} |\zeta(\frac{1}{2} + it)|^2 dt = \int_0^T |\zeta(\frac{1}{2} + it)|^2 dt, \quad (27)$$

the outer expectation being trivial since  $|\zeta(\frac{1}{2} + it)|^2$  is not a chance quantity. On the right,  $\Theta(T) = \theta(T) + cT = \frac{T}{2} \log(T/(2\pi)) - \frac{T}{2} + cT + O(1)$ . Hardy–Littlewood ([11], §7.3) gives  $\int_0^T |\zeta(\frac{1}{2} + it)|^2 dt = T \log T - (1 + \log 2\pi - 2\gamma)T + O(T^{1/2} \log T)$ . Dividing both (26) and the Hardy–Littlewood identity

by  $T \log T$  and taking  $T \rightarrow \infty$  equates the two leading coefficients: the right side of (26) contributes  $\sigma_Y^2 \cdot \frac{1}{2}$  and Hardy–Littlewood contributes 1, forcing

$$\sigma_Y^2 = 2. \quad (28)$$

No degree of freedom remains;  $\sigma_Y^2 = 2$  is a theorem derived from Hardy–Littlewood, not a convention. With  $\sigma_Y^2 = 2$ , (25) becomes the exact identity

$$\text{Var } Z(t) = 2 \Theta'(t), \quad (29)$$

with  $\Theta'(t) = \theta'(t) + c = \frac{1}{2} \text{Re } \psi(\frac{1}{4} + \frac{it}{2}) - \frac{1}{2} \log \pi + c$  by (??). No asymptotic expansion of  $\theta'$  enters: (29) holds for every  $t \in \mathbb{R}$  as an identity between  $\text{Var } Z$  and the digamma function.

## A.10 Sample-path autocorrelation under the unitary time change

The stationary process  $Y$  carries a sample-path autocorrelation, and the unitary time change expresses it directly on  $Z$ . For any window  $[A, B] \subset \mathbb{R}_u$  in the  $u$ -variable and any lag  $\eta \in \mathbb{R}$ ,

$$C_Y(\eta; A, B) := \int_A^{B-\eta} Y(u) \overline{Y(u+\eta)} du \quad (30)$$

is the unnormalized sample autocorrelation of  $Y$  on  $[A, B]$  at lag  $\eta$ . Apply the substitution  $u = \Theta(\tau)$ ,  $du = \Theta'(\tau) d\tau$ , using  $Y(u) = Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$  from (??). The pair  $(u, u + \eta)$  becomes  $(\Theta(\tau), \Theta(\tau) + \eta)$ , and the second argument is the image of  $\Theta^{-1}(\Theta(\tau) + \eta)$  under  $\Theta$ . Writing  $\tau'(\tau, \eta) := \Theta^{-1}(\Theta(\tau) + \eta)$  for the lagged time in  $t$ -coordinates,

$$C_Y(\eta; A, B) = \int_{\Theta^{-1}(A)}^{\Theta^{-1}(B-\eta)} Z(\tau) \overline{Z(\tau'(\tau, \eta))} \sqrt{\frac{\Theta'(\tau)}{\Theta'(\tau'(\tau, \eta))}} d\tau \quad (31)$$

where the integrand combines the two half-Jacobian factors  $1/\sqrt{\Theta'(\tau)} \cdot 1/\sqrt{\Theta'(\tau'(\tau, \eta))}$  from the two evaluations of  $Y$  with the single Jacobian  $\Theta'(\tau)$  from  $du = \Theta'(\tau) d\tau$ , giving the asymmetric ratio  $\sqrt{\Theta'(\tau)/\Theta'(\tau'(\tau, \eta))}$ . Equation (31) is the  $t$ -variable expression of the stationary sample-path autocorrelation of  $Y$  at lag  $\eta$ : translation-invariant on the  $u$ -axis becomes non-translation-invariant on the  $t$ -axis precisely through the asymmetric ratio and the  $\Theta^{-1}$ -images of the window endpoints. At zero lag,  $\tau'(\tau, 0) = \tau$ , the ratio is 1, and

$$C_Y(0; A, B) = \int_A^B |Y(u)|^2 du = \int_{\Theta^{-1}(A)}^{\Theta^{-1}(B)} |Z(\tau)|^2 d\tau = \int_{\Theta^{-1}(A)}^{\Theta^{-1}(B)} \left| \zeta\left(\frac{1}{2} + i\tau\right) \right|^2 d\tau \quad (32)$$

using  $|Z(\tau)| = |\zeta(\frac{1}{2} + i\tau)|$  on the real line. The right-hand side is the integrated second moment of  $\zeta$  on the critical line over the window  $[\Theta^{-1}(A), \Theta^{-1}(B)]$ ; as the window expands, it returns the Hardy–Littlewood asymptotic  $T \log T$  and couples directly to (25) through  $\sigma_Y^2 = 2$ . Identity (31) is independent of  $c$  only up to the  $c$ -dependence of  $\Theta$ : changing  $c$  rescales the window in  $t$ -coordinates through  $\Theta^{-1}$  but does not change the integrated identity (32).

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